UE 7 : Mobility and Smart Cities

part IV (last)

Philippe Canalda

Philippe.Canalda@univ-fcomte.fr

Remember …

- 2 main objectives, a first one which is real-time organization of mobility (optimization, prediction), a second one which is continuity of positionning and mobility (indoor <-> outdoor) if it remains enough time (now we know we do not have this time. However the MSC project has consisted in the co-specification and implementation of « a smart application of a indoor-outdoor instantaneous group's rendez-vous navigation system ».

Part 4 is :

- 1h15 of course composed of :
  - [a presentation of carpooling with transhipments finally totally solved (finalized)] optionnal
  - the final part of course with specific elements in a – graph+sub-graph decomposition, graph properties, and the walk definition in depth and in width of a graph.
- 0h15 of practical work to evaluate the making changes algorithms + transitive closure and to finalize the exercices about special items
- 0h15 of MSC project

- Possible to planify 1h next week or after to help finalize :
  - Mini-project delivery,
  - And practical work delivery.

Your feeling ?

Final Dynamic Carpooling Intra-Modal with Transhipment(s).

- Illustrated and presented by Isham and Gautham to ROADEF 2021

Continuation of MSC project

- f1 pursued
  - It is Dijkstra's algorithm that calculates the shortest paths from any point to any point.
    Warning ! There is a degree of difficulty between isTransitive and this Transitive closure. During a complete walk over the matrix roll, if a modification occurs, and therefore an addition of edge is issued from the transitivity computation, it is necessary to redo a complete walk, in search of an improvement of the transitive closure. We stop when there has been no improvement or modification during the complete walk over the matrix roll.

Continuation of MSC project

- f2 :
  - The f2 function uses a list of people positions. One then determines the point within the list of all the (discrete) positions of the problem by calculating the total value of all the displacements of the people to converge towards this point.

    We keep the point that minimizes the overall movements of the people of the group of students using the application.
    In order to, then, be able to generate the routes of each one, it is sufficient to modify the time matrix into a matrix of cells. If the cell is null, it matches -1. Otherwise, the cell will be made up of a cost in time, or in km, or a more complex table depending on the mode (bike, foot, car, etc.) and the time of day for taking congestion into account.

Continuation of MSC project

- f2 pursued:
  - … This cell is also made up of the route in text format which explains how to go from i to j. Going from i to k through j will correspond to the concatenation of the route t [i] [j] with that t [j] [k]. And that”s all folk!

    You have the innovative elements that many mobility solutions have not yet implemented;)

Subject of practical work n ° 4 (or homework)

- The challenge of the TD work to follow this presentation session will be :
  - for a given example with L={5, 2, 1, .5, .2, .1, .05} and m=12.35
  - greedy algorithm
    - » how much execution time ? How much print screen memory is used ?
  - exhaustive and enumerated version algorithm
    - » how much execution time ? How much print screen memory is used ? How much dynamic memory is used to keep in mind the best solution(s) ?
  - cut and price and share version
    - » how much execution time ? How much print screen memory is used ? How much dynamic memory is used to keep in mind the best solution(s) ? How much we have shared dynamic memory ? How much cuts are done ?

Exam question

- For a relation, a diagram or a graph, determining the properties of reflexivity, symmetry, transitivity, equivalence classes, this allows to reduce the starting data. It also helps to reduce the set of good solutions (targets). For each characteristic, it is necessary to be able to estimate the saving in quantity (size of the initial data, complexity saved by the algorithm, savings on the calculated solutions).
- For a given R relation, you need to formulate the new problems and evaluate the savings. We will have to deduct from these savings, possibly the additional calculation of the elements of an equivalence class, the calculation of the solutions resulting from a symmetry that we have taken into account to reduce the diagram of the starting data, etc.

The particular elements of an ordered set Example (3/3)

- Let X = {1, 2, 3, 5, 10, 20, 30}, be a set partially ordered by the relation x | y (x divides y).
  For the following two parts,
  - P_1 = {2,3,5,10} and P_2 = {2,5,10}
    - qualify them
      - the upper bound of P,
      - the lower bound of P,
      - the maximum elements
      - and the minimum elements,
      - the largest element of P,
      - the smallest element of P,
      - the u.b., the l.b.,
      - the universal element
      - and the null element.

- 1/ TD solve exercises precently listed,
- 2/ the complexity evaluation of making change
- 3/ adapt the transitive closure, or is-transitive or the best Mo-d which minimizes the path between 2 points of a graph,
  to
  remove the transitive edges of a relation that means that, for any t[x][z], having t[x][y] and t[y][z], remove t[x][z] from the relation
- Bla-bla-bla

Part 4

The particular elements of an ordered set Example (3/3) first result

- Let X = {1, 2, 3, 5, 10, 20, 30}, be a set partially ordered by the relation x | y (x divides y).
  For the following two parts,
  - P_1 = {2,3,5,10}
    - qualify them
      - the upper bound of P, {30}
      - the lower bound of P, {1}
      - the maximum elements, {3, 10}
      - and the minimum elements, {2, 3, 5}
      - the largest element of P, inexistant
      - the smallest element of P, inexistant
      - the supremum, {30}
      - the infinimum, {1}
      - the universal element, inexistant
      - and the null element, {1}.

The particular elements of an ordered set Example (3/3) second result

- Let X = {1, 2, 3, 5, 10, 20, 30}, be a set partially ordered by the relation x | y (x divides y).
  For the following two parts,
  - P_2 = {2, 5,10}
    - qualify them
      - the upper bound of P, {10, 20, 30}
      - the lower bound of P, {1}
      - the maximum elements, {10}
      - and the minimum elements, {2, 5}
      - the largest element of P, {10}
      - the smallest element of P, inexistant
      - the supremum, {10}
      - the infinimum {1}
      - the universal element, inexistant
      - and the null element, {1}.

Use cases of these particular items 1/2

- The **a priori** knowledge of the data on which a combinatorial calculation (or optimizations) must be carried out.
  - case where the use of calculation clearly follows the availability of data over time ...
    - => so-called static optimizations where the result can be stored in the form of arrays, sagittal diagram / graph / automaton, Hass diagram, etc.
      This static part can sometimes correspond to the generation of data and / or code which, once compiled and added to a dynamic optimization kernel, will produce the expected results after compilation and then execution (see below).
  - ... and where sometimes the **a posteriori** use limits the scope of the combinatorial calculus on the initial data => so-called dynamic optimizations which will perhaps exploit the static processing prior to, and not necessarily exclusively,
    - Compilation, filtering, automata browsing or graph browsing or ...

Use cases of these particular items 2/2a

- On a given network or territory, we have the distances separating nodes 2 to 2.
  First, and for the sake of simplification, we will count the number of segments / edges separating different points / nodes (of a graph)

  G = (N = {0,1,2,3,4,5,6,7,8}, E = {(0.5), (0.7), (5.7), (5.3) , (3.6), (3.2), (7.3), (7.8), (7.1), (7.4), (4.8), (5.1)})

  We want to write a program which, given a subset of these nodes and edges, computes the list of nodes ordered in ascending order of distances to a given particular node.

  - For node 0 and the subset N0 = {1,3,4,5,7} and the associated edges E0 = {(0,5), (0,7), (5,7), (7, 3), (7,1), (7,4), (5,1)}
    and the expected result is L0 = {0,5,7,1,3,4}

Use cases of these particular items 2/2b

- G = (N = {0,1,2,3,4,5,6,7,8}, E = {(0.5), (0.7), (5.7), (5.3) , (3.6), (3.2), (7.3), (7.8), (7.1), (7.4), (4.8), (5.1)})
  - » For node 0 and the subset N0 = {1,3,4,5,7} and the associated edges E0 = {(0,5), (0,7), (5,7), (7, 3), (7,1), (7,4), (5,1)} and the expected result is L0 = {0,5,7,1,3,4}
  - This involves calculating the shortest paths from 0 to all points (by extension from any point to any point),
  - to arrange in L the nodes according to the increasing distance (but also in alphanumeric order in the event of equality _ because we have a partial order)
  - then during dynamic optimization, one have to browse the initial list L by showing the indexes of the nodes present in N0.

Exemples d'usages de ces éléments particuliers 2/2 c

- It's about
  - calculate the shortest paths from 0 to all points (by extension from any point to any point),
  - to arrange in L the nodes according to the increasing distance (but also in alphanumeric order in the event of equality _ because we have a partial order)
  - then during dynamic optimization, it suffices to browse the initial list L by showing the indexes of the nodes present in N0.
- Give the matrix representation,
- Characterize the relationship (properties)
- Give the sagittal representation
- Does the representation of the Hasse diagram make sense in the present case and why
- Design the algorithm i) which calculates the shortest paths from any point to 0
- Extend this algo into an algo ii) which returns the list ordered totally according to the distances then, in case of equality, the list ordered according to alphanumeric order
- Design the algorithm iii) which calculates the sublist L0 corresponding to the subgraph G0 = (N0, EO) = G \ {2,4,6,8}
- Design the algorithm which extends i) to iv) which calculates the shortest paths from any point to any point
- Discussion VERY INTERESTING EXERCISE TO PERFORM

Trellises - definition 1/2

- When the part P of X is reduced to 2 elements ("pair") {x, y}, there may exist a ub. and / or a l.b.
  - The u.b. when it exists, denoted by x V y
  - The l.b. when it exists, denoted by x Λ y

- Definition: We call a lattice a partially ordered set in which, for any pair of elements, there exists a u.b. and a l.b.

Trellises - definition 2/2

- Let T be a set, whose elements are endowed with 2 laws of composition, V (OR) and Λ (AND), verifying, whatever is x, y and z Є T, the properties:
  - 1'- x OR y = y OR x (commutativity) 1- x AND y = y AND x
  - 2'- x OR (y OR z) = (x OR y) OR z (associativity) 2- x AND (y AND z) = (x AND y) AND z
  - 3'- x OR x = x (idempotence) 3- x AND x = x
  - 4'- x OR (x AND y) = x (absorption) 4- x AND (x OR y) = x
- Then T constitutes a set ordered by the relation ≤ such that:
  - 5- [x ≤ y] ⇔ [x AND y] = x ⇔ [x OR y] = y
- T is a treillis

Trellises – example

- Consider the set T of the divisors of 30: 1, 2, 3, 5, 6, 10, 15, 30, ordered by the relation x | y.
- If we consider the associated Hasse diagram (and which you will construct),
- and if we consider that the u.b. of 2 elements is their scm and that the l.b. of 2 elements is their gcd,
- Then T, associated with l.b. and u.b. form a lattice

- The universal element is 30, the null element is 1.
- In general, N * ordered by '|' presents an infinite lattice structure whose the null element is always 1 and not having a universal element
- Develop the 2 Hass diagrams, the one for x | y, then the new relation

Trellises – other definitions 1/2

- A trellis T with zero element ***n*** and universal element ***U*** is complemented if, whatever x of T, there exists a possible association of an element x denoted by x-bar such that:
  - 6- x OR x-bar = U and 6'- x AND x-bar = n
- Exercise
  - check if the trellis of the previous example is complemented
  - Show that the system of axioms (1-4), (1'-4 ') is not minimal, especially you will be able to show that absorption leads to idem-potence.
    Thus the system of axioms (1, 2, 4, 1 ', 2', 4 ') is minimal or generator.

Trellises – other definitions 2/2

- A lattice is distributive if, to axioms 1 to 4 and 1 ’to 4’ are added, some x, y and z of T:
  - 7- x OR (y AND z) = (x OR y) AND (x OR z)

- Exercise
  - These axioms lead to :
    - 7’- x AND (y OR z) = (x AND y) OR (x AND z)
  - Conversely 7 'results in 7
  - In a distributive lattice the complementation is unique (demonstration by the absurd)

- A lattice that is both distributive and complemented is called a Boolean lattice. It is isomorphic to a Boolean algebra which is:
  - A partially ordered set, endowed with a null element and another universal, whose elements satisfy 1 to 4, 1 'to 4', 6, 6 ', 7
    The order relation is defined by one of the following 4 relations :
    - 5*- [x ≤ y] ⇔ [x AND y] = x ⇔ [x OR y] = y ⇔ x AND y-bar = 0 ⇔ x OR y-bar =1, with n=0 and U=1

A Boolean algebra

- A Boolean algebra with one generator, other than 0 and 1, has 4 elements. (Hass diagram)
- With 2 elements, it has 16 elements.

- Stone's theorem: Any Boolean algebra is isomorphic to a set field.

- Example: from Boolean algebra with 2 generators a and b such that a! = B and a AND b! = 0,
  we make it appear more clearly ...

Body of a set (corps d'ensemble)

- Example: from Boolean algebra with 2 generators a and b such that a! = B and a AND b! = 0,
  we make it appear more clearly…

- In a parallelepipedal shape (Euler-Venn diagram) representing universe 1: A, B, A ∩ B, A ∩ B-bar, A-bar ∩ B, A-bar ∩ B-bar which are "regions" corresponding to the respective atoms a, b, a AND b, a OR b-bar, a-bar AND b, a-bar AND b-bar

- We call a set body a family of parts of a set, endowed with classical set operations
  (union, intersection, complementation).

Reminder on graphs

- Oriented graph G
  - A finite set V of vertices x_1, … x_n
  - A family E of items from the cartesian product N x N, called edges

    an edge can appear several time
- An oriented edge (x,y) and a non oriented one (symmetrical) xy

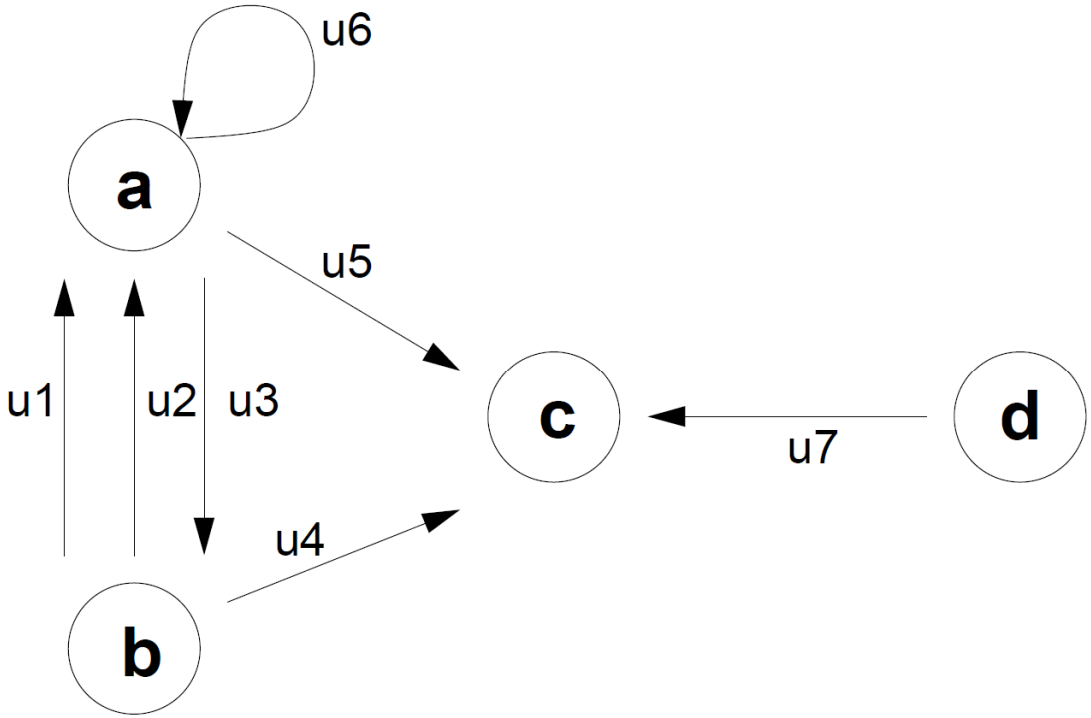
slidesGraphes_in_french_PhC-part5.pdf

- Éléments de graphes en français téléchargeable depuis discord

MIT6_042JF10_chap05-part5.pdf

- Long version english support established by MIT

Reminder on graphs

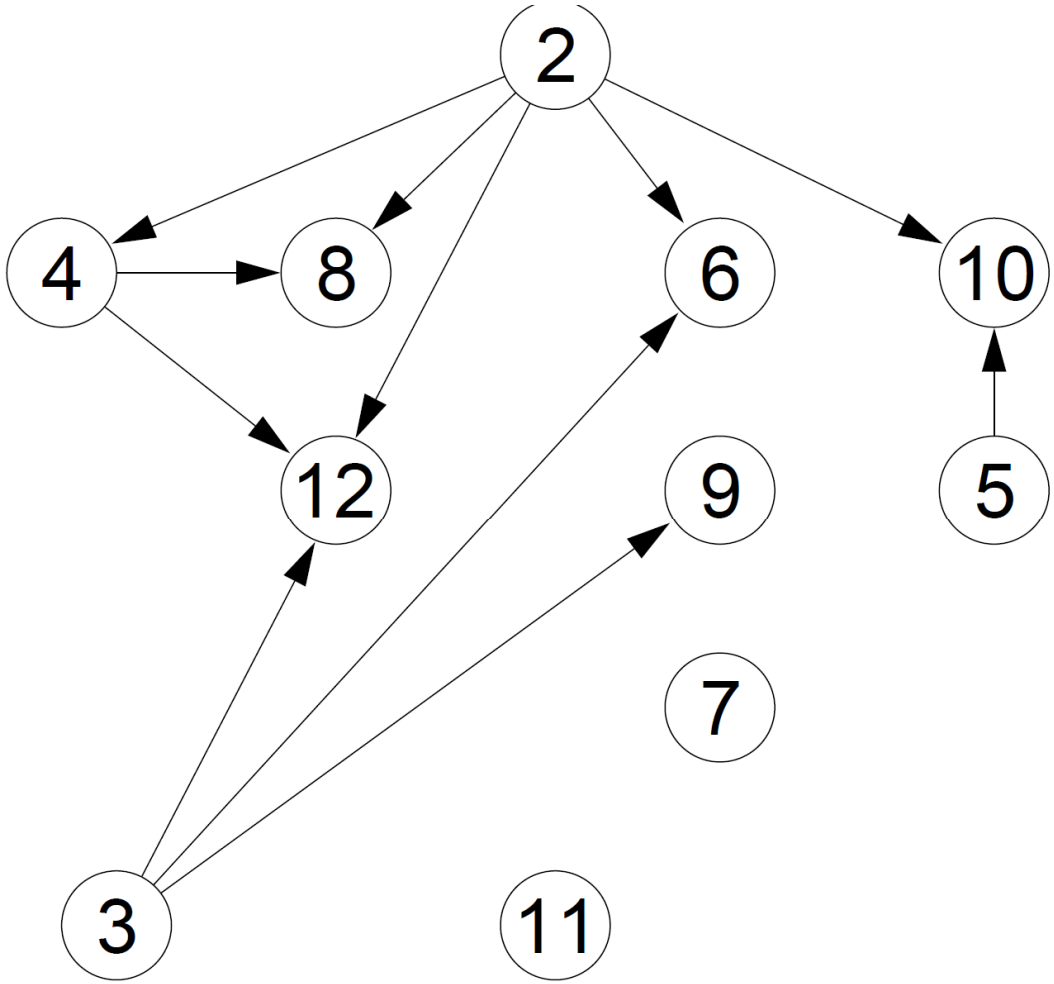


- A 2-graph, a p-graph

Reminder on graphs

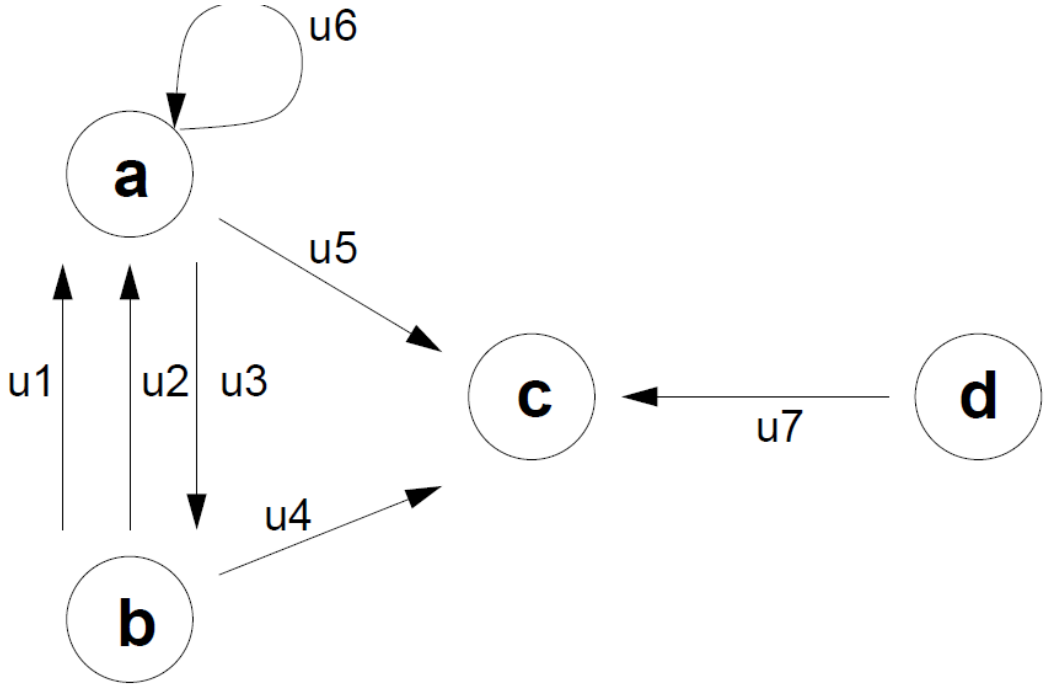
- Relation between nodes
  - Symmetrical, not symmetrical, transitive, etc.

Reminder on graphs



An example :
the relationship x divides exactly y

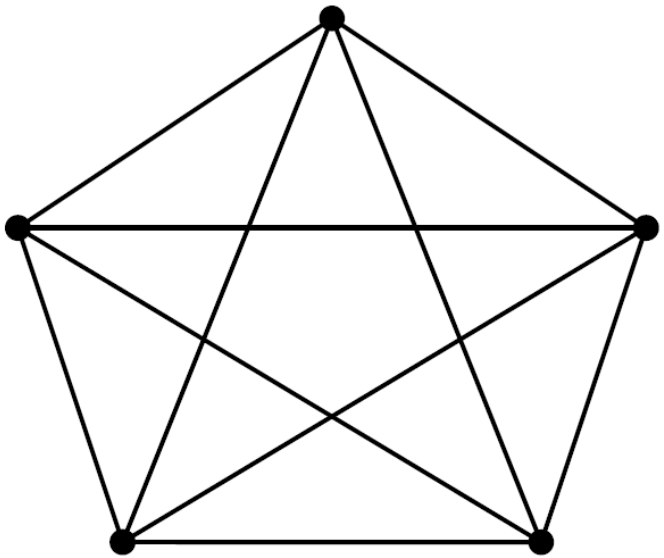
Reminder on graphs



- saggital diagram and weighted graph
- Incident edge, joining a vertex / node
- Self-loop and Loop
- Degree, subgraphs, adjacency matrix,

Reminder on graphs

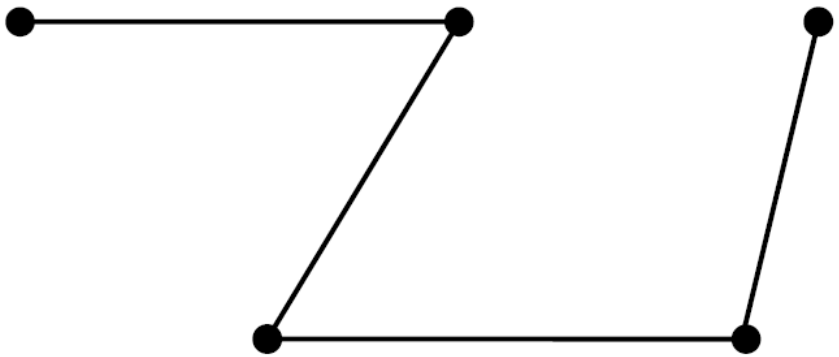
- Common graphes, and simple one's



- The complete 1-graph on n vertices, denoted K_n has an edge between every 2 vertices. n (n-1) / 2 edges

Reminder on graphs

- Common graph is for example, the 5-node line graph L5



- An 1-graph with n-1 edges in sequence
- Complete graph
- Hypercube Hn
- Cube Connected Cycle CCCn
- Grid, toric grid, buterfly graph, shuffle exhange, star fly, …

Reminder on graphs

- Simplest chain, shortest path, a loop, a graph without loop, a directed acyclic graph (DAG), incident edge(s), neighbor(s), adjacent vertex/node(s), degree (od a node, of a graph), density of a graph, …
- Particular graphs :
  - regular graph, $\delta(G) = \Delta(G)$
  - irregular's one,
  - Symmetrical graph, anti-symetrical one
  - Complementary graph
  - Inverse graph
- Diameter and distances : d(x,y), d(xy,uv), d(L1,L2), …

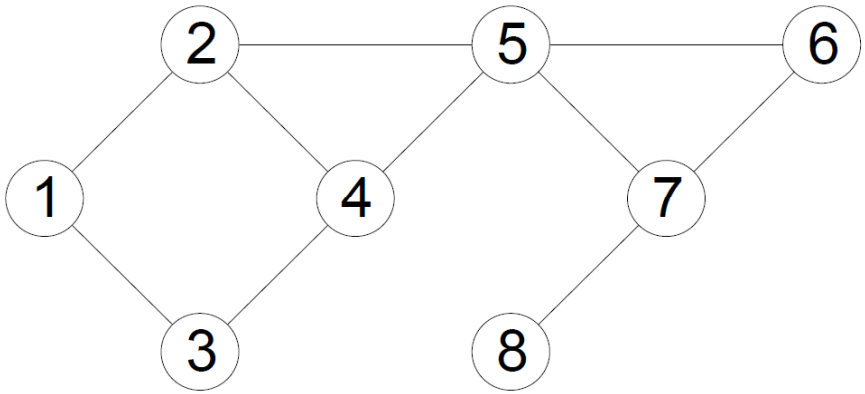
Reminder on graphs

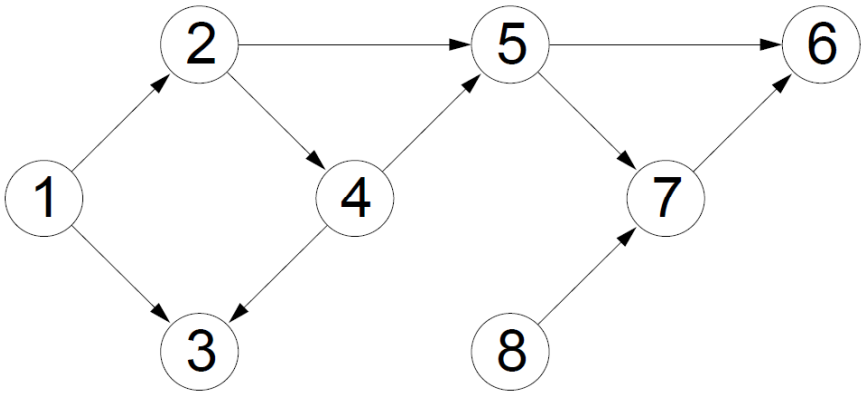
- Operations on graphs:
  - union
  - intersection
- Complete graph, isomorphism and bipartite graph
- Searching a complete sub-graph in a graph is a NP-complete problem

Reminder on graphs

- Path, elementary
  path, circuit,

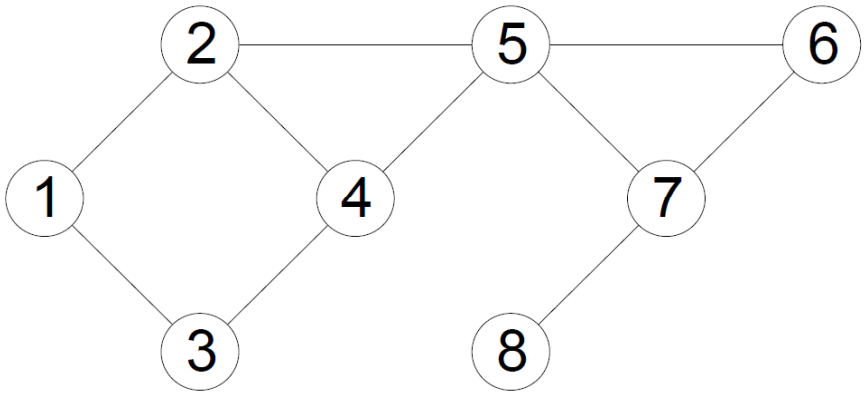
Generic browse : border

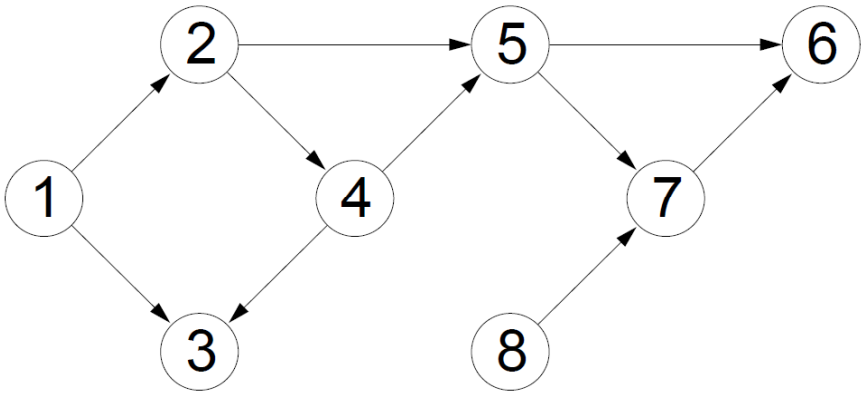




- G(V,E) with |V|=n and |E|=m, and a set $T \subset V$
- Border of T, B(T,G), is a set of node(s) of V-T, node which is at least a neighbour of one node of T

Generic browse : permutation



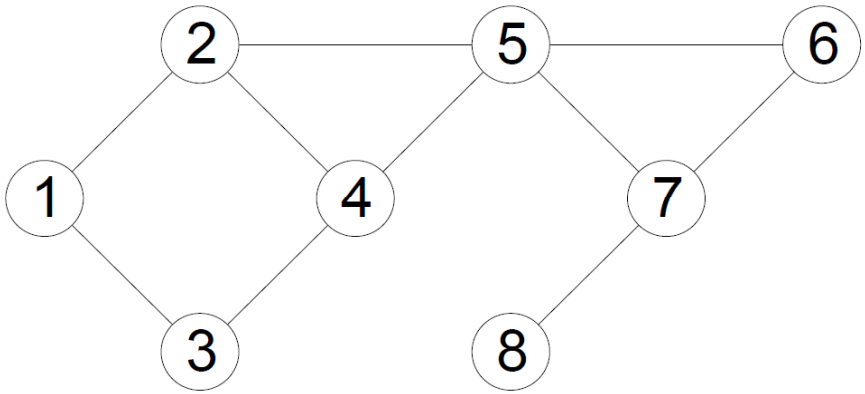


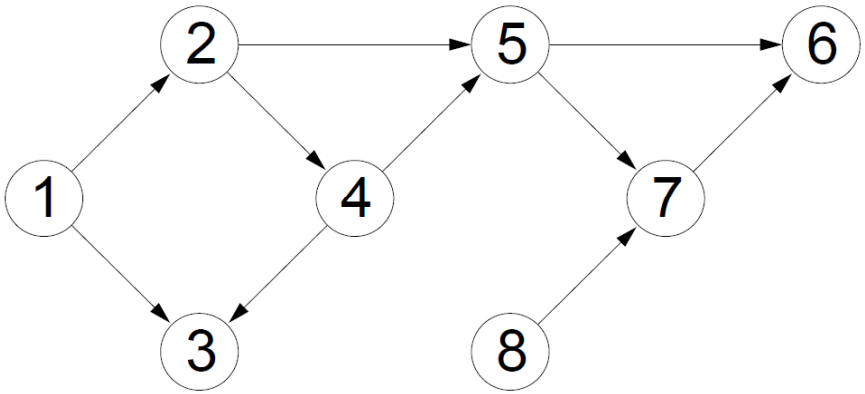
- A permutation L=(S1, S2 .., Sn) of nodes from a graph G a generic browse from the node i if and only if:

$$\forall j \in \{2, \ldots, n\},\ B(L[1, j-1], G) \neq \emptyset \Rightarrow s_j \in B(L[1, j-1], G)$$

with L[1,j-1] a sub-list of nodes S_1, .. S_{j-1}

Generic browse : permutation



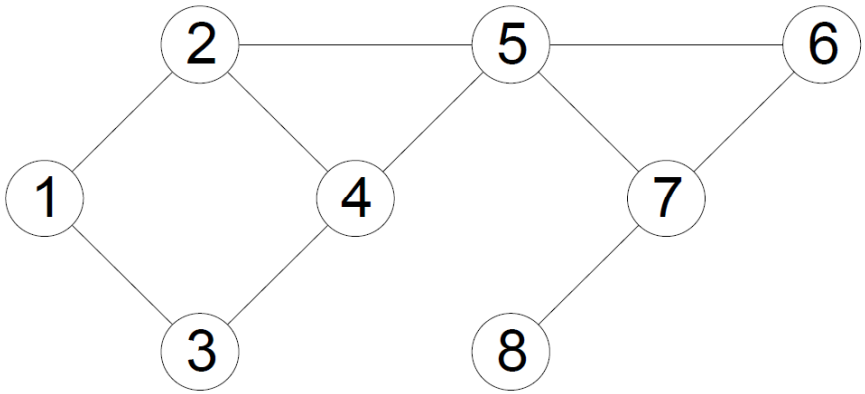


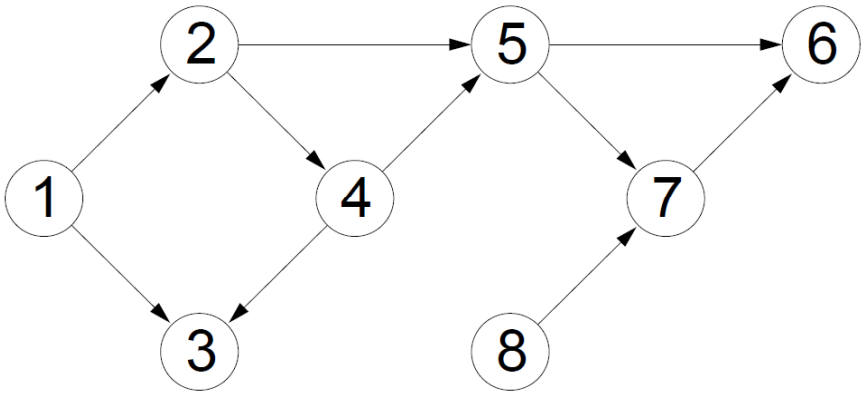
- A permutation L=(S1, S2 .., Sn) of nodes from a graph G a generic browse from the node i if and only if:

$$\forall j \in \{2, \ldots, n\},\ B(L[1, j-1], G) \neq \emptyset \Rightarrow s_j \in B(L[1, j-1], G)$$

with L[1,j-1] a sub-list of nodes S_1, .. S_{j-1}
? L1 = (1, 3, 4, 5, 7, 8, 6) ?

Generic browse : permutation



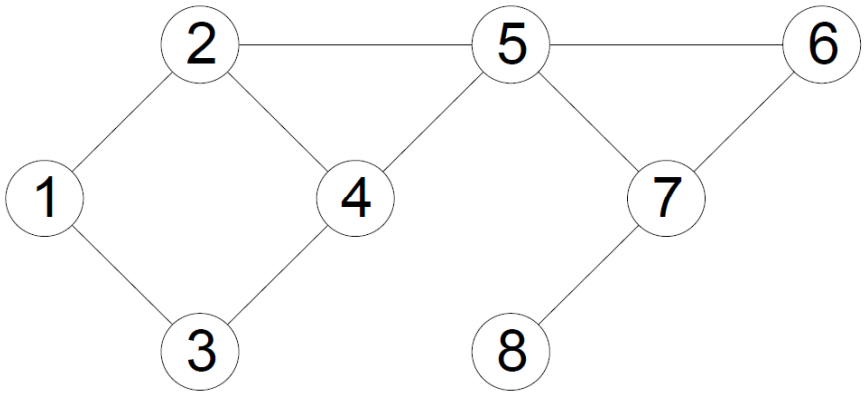


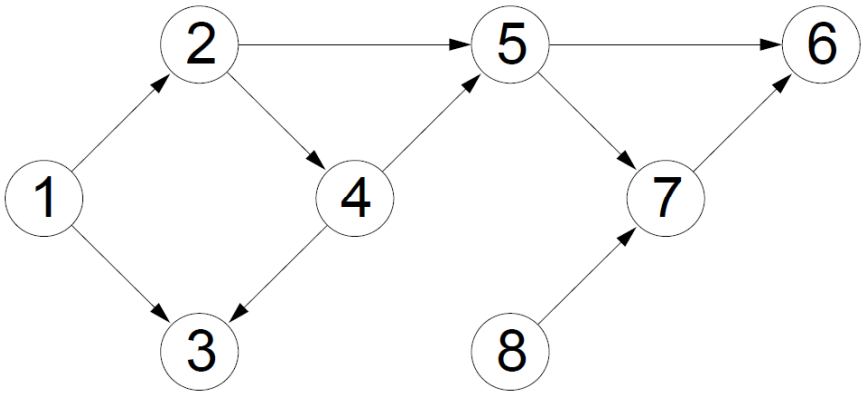
- A permutation L=(S1, S2 .., Sn) of nodes from a graph G a generic browse from the node i if and only if:

$$\forall j \in \{2, \ldots, n\},\ B(L[1, j-1], G) \neq \emptyset \Rightarrow s_j \in B(L[1, j-1], G)$$

with L[1,j-1] a sub-list of nodes S_1, .. S_{j-1}
? L1 = (1, 3, 4, 5, 7, 8, 6) ? Is a possible browse

Generic browse : permutation



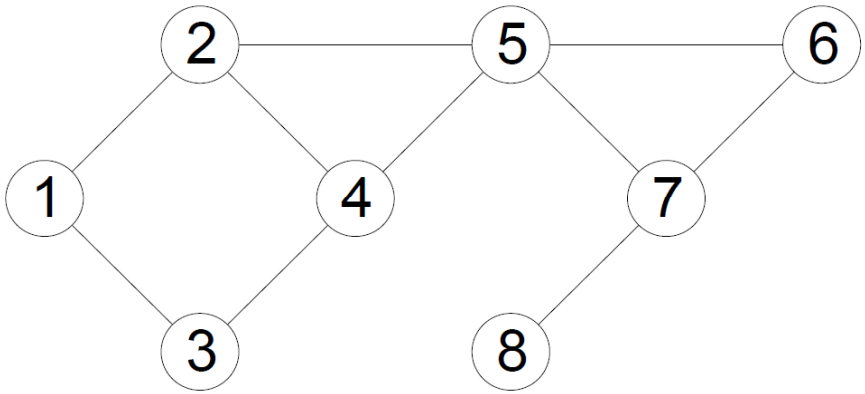


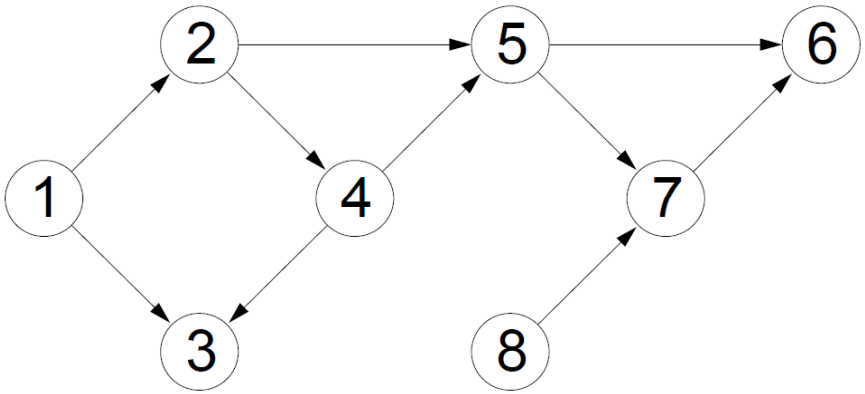
- A permutation L=(S1, S2 .., Sn) of nodes from a graph G a generic browse from the node i if and only if:

$$\forall j \in \{2, \ldots, n\},\ B(L[1, j-1], G) \neq \emptyset \Rightarrow s_j \in B(L[1, j-1], G)$$

with L[1,j-1] a sub-list of nodes S_1, .. S_{j-1}
? L2 = (5, 6, 2, 7, 8, 1, 3, 4) ?

Generic browse : permutation



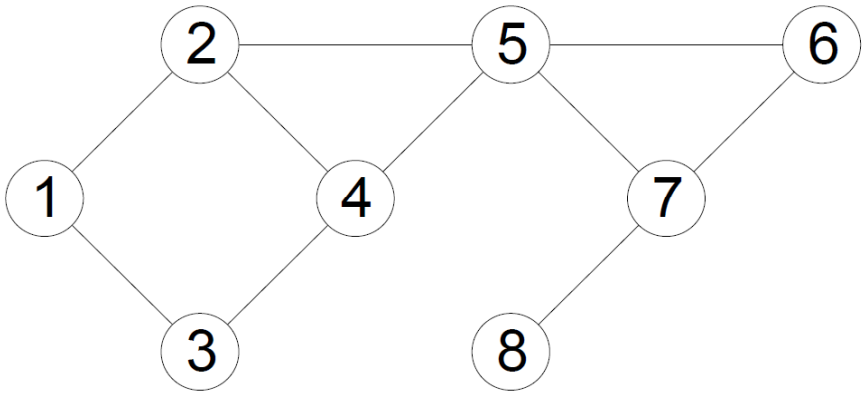


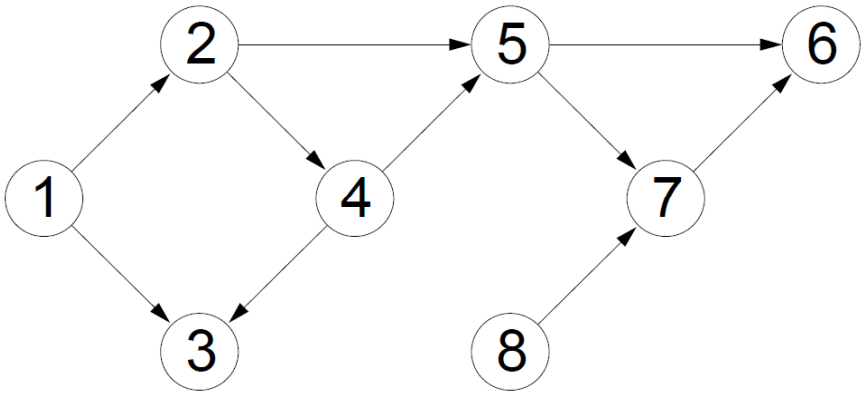
- A permutation L=(S1, S2 .., Sn) of nodes from a graph G a generic browse from the node i if and only if:

$$\forall j \in \{2, \ldots, n\},\ B(L[1, j-1], G) \neq \emptyset \Rightarrow s_j \in B(L[1, j-1], G)$$

with L[1,j-1] a sub-list of nodes S_1, .. S_{j-1}
? L2 = (5, 6, 2, 7, 8, 1, 3, 4) ? Is a possible browse

Generic browse : permutation



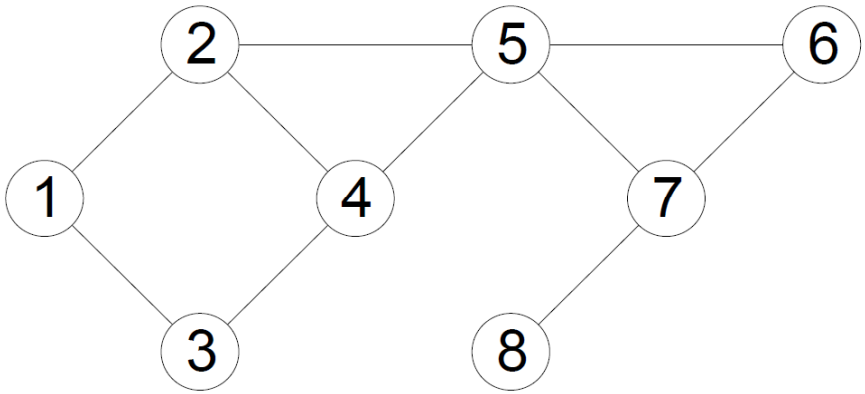


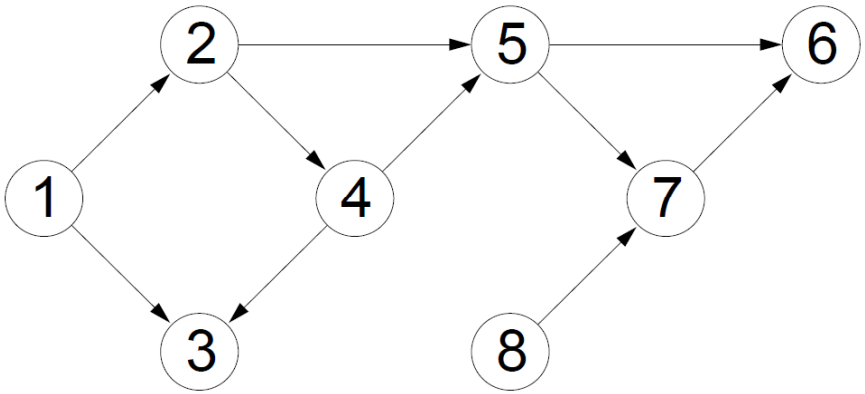
- A permutation L=(S1, S2 .., Sn) of nodes from a graph G a generic browse from the node i if and only if:

$$\forall j \in \{2, \ldots, n\},\ B(L[1, j-1], G) \neq \emptyset \Rightarrow s_j \in B(L[1, j-1], G)$$

with L[1,j-1] a sub-list of nodes S_1, .. S_{j-1}
? L3 = (5, 6, 3, 2, 1, 4, 8, 7) ?

Generic browse : permutation





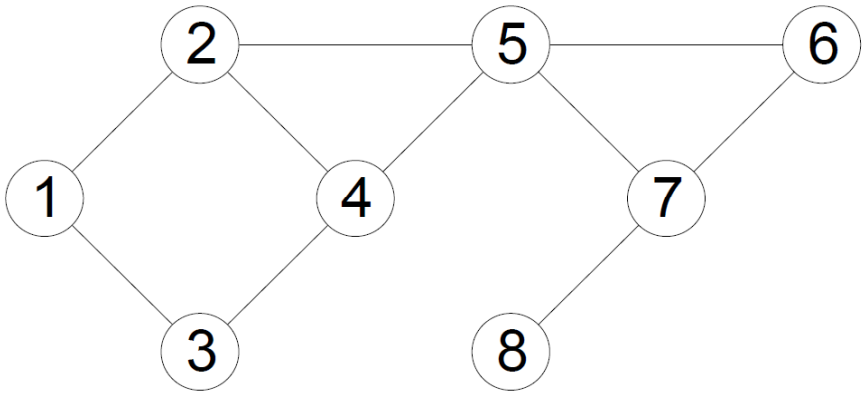
- A permutation L=(S1, S2 .., Sn) of nodes from a graph G a generic browse from the node i if and only if:

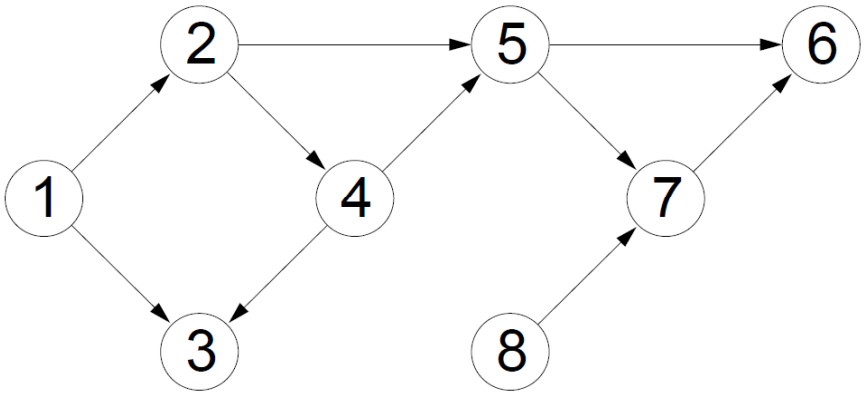
$$\forall j \in \{2, \ldots, n\},\ B(L[1, j-1], G) \neq \emptyset \Rightarrow s_j \in B(L[1, j-1], G)$$

with L[1,j-1] a sub-list of nodes S_1, .. S_{j-1}

? L3 = (5, 6, 3, 2, 1, 4, 8, 7) ? No $s_3 \notin B(L_3[1, 2])$.

Generic browse : regeneration point





- Given L of G, a regeneration point of L is a node Sk such that k=1

  or B (L[1, k-1],G) = ∅ with k > 1

  these points are initiators, or null points, helpfull in multiple wave browse of a graph.

Choice graph, covering forest

- Given L of G, foreach node Sj not a regeneration node, we bind it to one (Si,Sj) of G such that $s_i \in L[1, j-1]$.

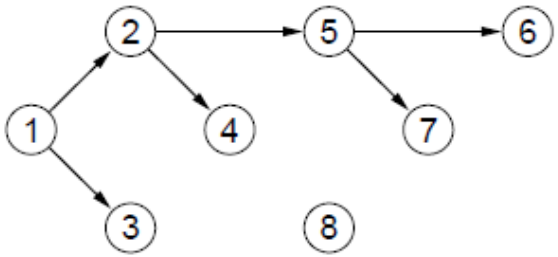
  The obtained graph G' is called choice graph or covering forest

Choice graph, covering forest

- Given L of G, foreach node Sj not a regeneration node, we bind it to one (Si,Sj) of G such that $s_i \in L[1, j-1]$.

  The obtained graph G' is called choice graph or covering forest

  L4=(1, 2, 3, 4, 5, 6, 7, 8) is a generic browse of G



  is one choice graph of G and it is not unic ((4,5) instead of (2,5))

  The node 8 is a regeneration node

Opened node, closed node

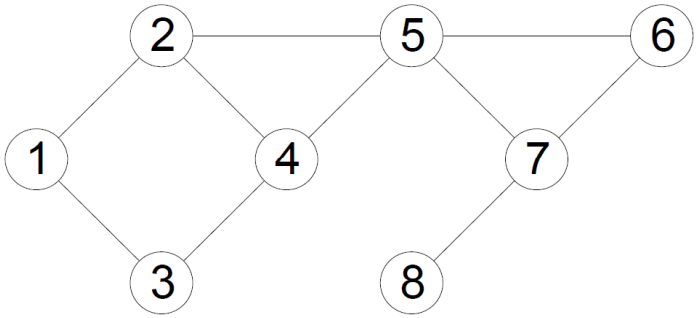
- A node x of a permutation L, which is a generic browse L[1,i] = (s1, .. Si), is opened in L[1, i] for G if it accepts at least one neighbour in G in L[i+1, n]. Otherwise, this node is closed in L[1,i] for G.

In width walk/browse

- A generic browse L = (s1, …, Sn) is in width if
  foreach j in {2, …, n} such that B(L[1, j-1],G)≠∅
  the node sj is a neighbour of the first opened node for L[1, j-1] in G.

In width walk/browse

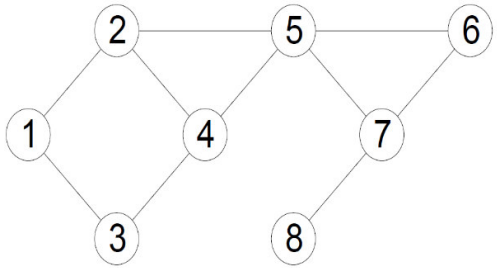
- A generic browse L = (s1, …, Sn) is in width if
  foreach j in {2, …, n} such that B(L[1, j-1],G)≠∅
  the node sj is a neighbour of the first opened node for L[1, j-1] in G.



? L5 = (1, 2, 3, 4, 5, 6, 7, 8)
and L6 = (6, 7, 5, 8, 4, 2, 3, 1)
and L6bis = (6, 5, 7, 8, 4, 2, 3, 1) ?

In width walk/browse

- A generic browse L = (s1, …, Sn) is in width
  if
  foreach j in {2, …, n} such that B(L[1, j-1],G)≠∅
  the node sj is a neighbour of the first opened node
  for L[1, j-1] in G.



? L5 = (1, 2, 3, 4, 5, 6, 7, 8) in with
and L6 = (6, 7, 5, 8, 4, 2, 3, 1) in width
and L6bis = (6, 5, 7, 8, 4, 2, 3, 1) ? not in width
because s4 = 8 is not adjacent to the first opened node for L[1, 3] in G (node 5)

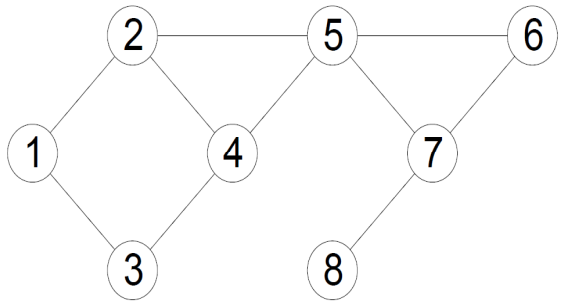
In depth walk/browse

- A generic browse L = (s1, …, Sn) is in depth
  if
  foreach j in {2, …, n} such that B(L[1, j-1],G)≠∅
  the node sj is a neighbour of the last opened node
  for L[1, j-1] in G.

In depth walk/browse

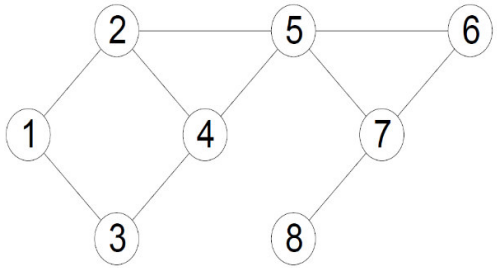
- A generic browse L = (s1, …, Sn) is in depth
  if
  foreach j in {2, …, n} such that B(L[1, j-1],G)≠∅
  the node sj is a neighbour of the last opened node
  for L[1, j-1] in G.

  ? L7 = (2, 5, 4, 3, 1, 6, 7, 8)
  and L8 = (5, 7, 8, 2, 4, 1, 3, 6) ?



In depth walk/browse

- A generic browse L = (s1, …, Sn) is in depth
  if
  foreach j in {2, …, n} such that B(L[1, j-1],G)≠∅
  the node sj is a neighbour of the last opened node
  for L[1, j-1] in G.



? L7 = (2, 5, 4, 3, 1, 6, 7, 8) in depth
and L8 = (5, 7, 8, 2, 4, 1, 3, 6) not in depth
because s4 = 2 is not adjacent to the last opened node for L[1, 3] in G (node 7)

Reminder on graphs

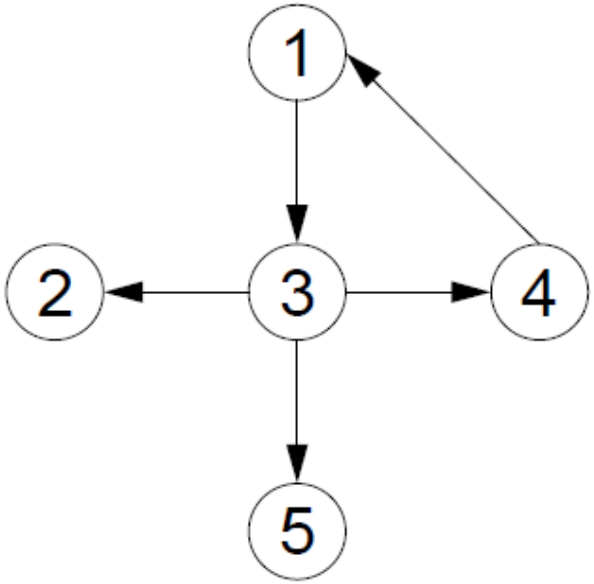
- Transitive closure

$$\Gamma^+(x) = \{x\} \cup \Gamma(x) \cup \Gamma(\Gamma(x)) \ldots$$

Reminder on graphs

- Transitive closure

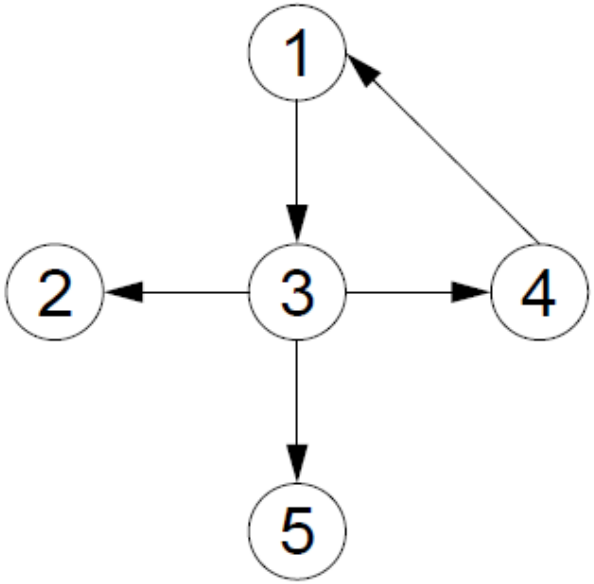
$$\Gamma^+(x) = \{x\} \cup \Gamma(x) \cup \Gamma(\Gamma(x)) \ldots$$

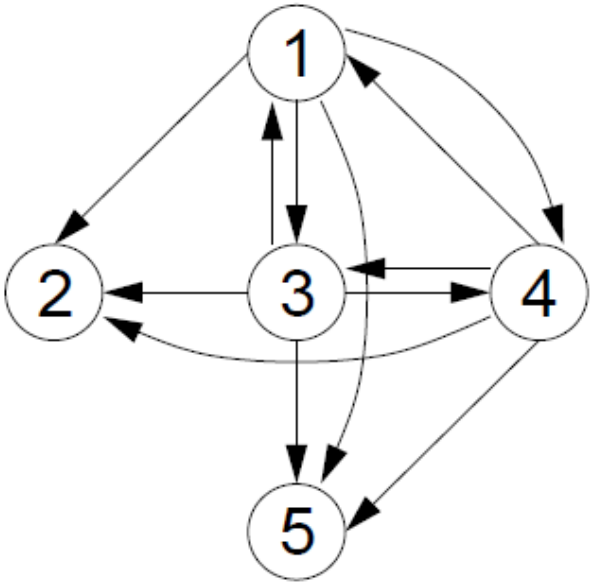


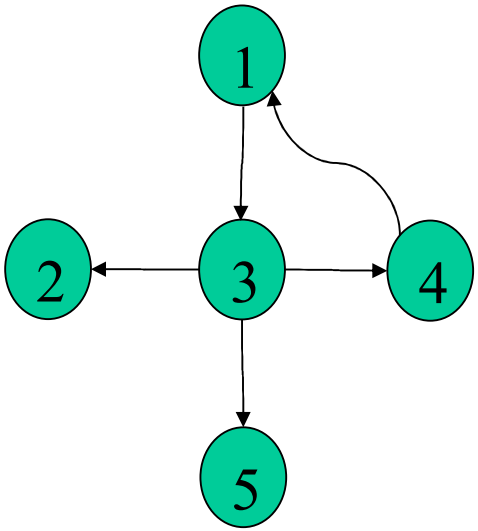
Reminder on graphs

- Transitive closure

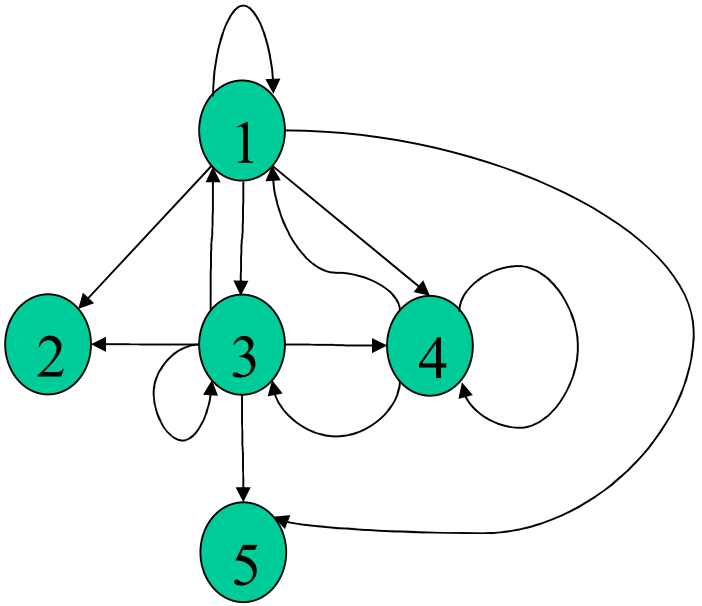
$$\Gamma^{+}(x) = \{x\} \cup \Gamma(x) \cup \Gamma(\Gamma(x)) \ldots$$







An oriented 1-graphe G *with an order of* 5



transitive closure of G

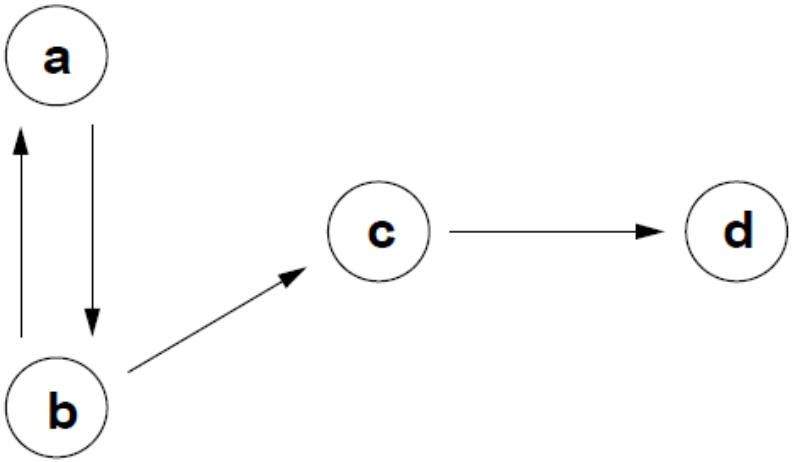
Reminder on graphs

- Computing of transitive closure of a graphe G
  - Given M the adjacency matrix of G

    the matrix M^+ of G^+, transitive closure of G is computed according to :
    M^+ = M + M^1 + … M^{n-1}

Reminder on graphs

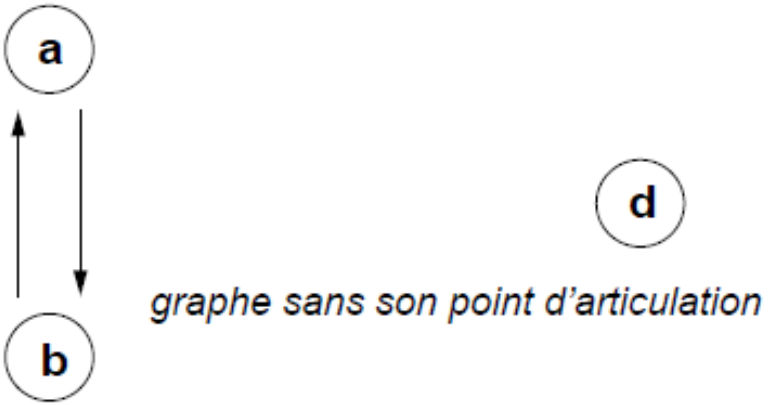
- Eulerian path, Hamiltonian path, …
- Connexity and strong connexity



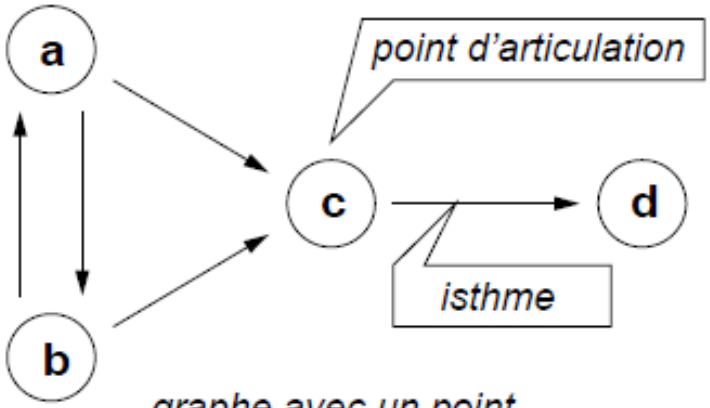
- isthmus and point of articulation

  – (if you build a stringly connected graphs from 10 locations (indoor, outdoor for example) and you compute the transitive closure, you obtain a complete graph, that means a matrix origine-to-destination describing all paths from any point of the territory to any other point of the same territory)

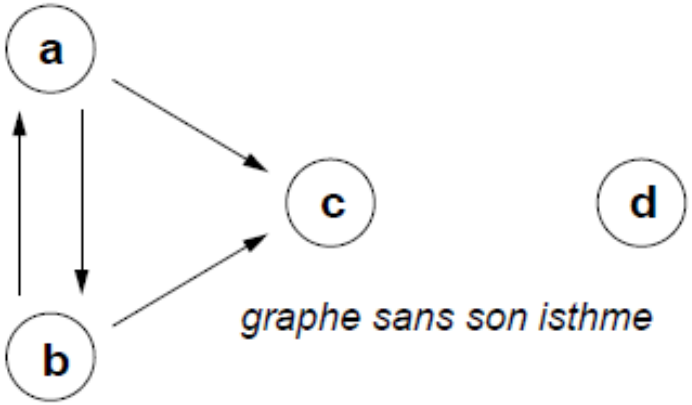
Reminder on graphs



graphe sans son point d'articulation



graphe avec un point articulation et un isthme



graphe sans son isthme

To come …

- Coming next :
  - 18th of november from 13h30 p.m. to 16h30
  - 27th of november from 14h p.m. to 16h exam 50 % of the mark
  - 1st delivery of practical works + conception of mini-project ASAP before 30th of november and a second version until the christmas holidays (20%)
  - Delivery of the project until the 16th of january (30%). But do it asap then you can get a feedback and a mark likely to be improved